

Technology #124

Incorporate electrical impedance tomography
sensors in venous filter to detect increased
pulsatility as an early indicator of flow
obstruction

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-30

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Incorporate electrical impedance tomography sensors in venous filter to detect increased pulsatility as an early indicator of flow obstruction

WHY DOES IT MATTER?

Documented recurring problem 'obstruction_of_flow' for venous filter: 9 MAUDE reports across 8 distinct devices (20161117 to 20260505); root-cause evidence from 0 recall records and 9 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 9 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, custody-attached; the aggregation across records is AI_INFERENCE (bounded by the re Unresolved gap: Across the 7 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, recurring, reported, un

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in ml_data. 6-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Compare filter obstruction rates with and without pulsatility monitoring in a controlled clinical study. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: held-out + temporal validation splits. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data; empirical risk on a non-representative sample says nothing about deployment risk (distribution shift); utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured; ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.; ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.; ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.; Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII.); Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate [y_i , -] value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Regional pulsatility amplitude increases with downstream flow obstruction

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'obstruction_of_flow' for venous filter: 9 MAUDE reports across 8 distinct devices (20161117 to 20260505); root-cause evidence from 0 recall records and 9 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 9 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, custody-attached; the aggregation across records is AI_INFERENCE (bounded by the re Unresolved gap: Across the 7 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, recurring, reported, un

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: ml_data

Disciplines: data engineering, machine learning, software engineering, statistics

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: Incorporate electrical impedance tomography sensors in venous filter to detect increased pulsatility as an early indicator of flow obstruction

Subsystems:

- ?: data ingestion and labeling pipeline - ?
- ?: feature / representation layer - ?
- ?: model + uncertainty quantification - ?
- ?: deployment target + monitoring loop - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Incorporate electrical impedance tomography sensors in venous filter to detect increased pulsatility as an early indicator of flow obstruction

Key Physics: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: CONDITIONAL; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: NOT ESTABLISHED.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: CONDITIONAL; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: NOT ESTABLISHED.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: CONDITIONAL; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: NOT ESTABLISHED.

Assumptions:

- priors and likelihoods are MODEL_DERIVED until validated on deployment data
- empirical risk on a non-representative sample says nothing about deployment risk (distribution shift)
- utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured
- ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.
- ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.
- ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.
- Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.

- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate [y_i, -] value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'obstruction of flow', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'distribution shift degrading accuracy post deployment', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'label noise propagating to biased predictions', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'overfitting to source-site artifacts', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'calibration drift invalidating decision thresholds', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'feedback-loop data contamination', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Devices matching 'venous filter' must maintain function under the conditions of blood flow reduction during injury events with reported venous thrombotic events.	Goldberg, 2017
DI-002	Mechanism driver (proposed)	Regional pulsatility amplitude increases with downstream flow obstruction	MODELLED
DI-003	Expected effect target	Early detection of filter obstruction before complete blockage occurs, allowing timely intervention	MODELLED
DI-004	Domain design input pattern: decision applicability NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	Decision applicability NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING PROPOSED
DI-005	Domain design input pattern: available applicability NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	Available applicability NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING PROPOSED
DI-006	Domain design input pattern: latent/inapplicable NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	Latent/inapplicable NOT ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING PROPOSED
DI-007	Regulatory design input: compliance standards (e.g., FDA 21 CFR 801.404) Machine Learning Product Development Process	compliance standards (e.g., FDA 21 CFR 801.404) Machine Learning Product Development Process	FDA 21 CFR 801.404
DI-008	Custodied observation: Conceptual framework of the CTS (Crisis Management System) for emergency response	https://www.fda.gov/cdrh/oc/tpr/cts/	https://www.fda.gov/cdrh/oc/tpr/cts/
DI-009	Custodied observation: Pulmonary embolism identified by electrical impedance tomography	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/
DI-010	Custodied observation: Management of respiratory distress in patients with acute pulmonary embolism	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/
DI-011	Custodied observation: Spontaneous pneumothorax in patients with acute pulmonary embolism	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/
DI-012	Custodied observation: Experience treating hypoxemic patients with acute pulmonary embolism	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6885510/

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	data ingestion and labeling pipeline realizing Incorporate electrical impedance tomography sensors in definition	CONCEPTUAL	details by sensors in definition filter to detect increased pulsatility as an early indicator
DO-002	feature / representation layer realizing Incorporate electrical impedance tomography sensors in definition	CONCEPTUAL	details and invariance definition to detect increased pulsatility as an early indicator
DO-003	model + uncertainty quantification realizing Incorporate electrical impedance tomography sensors in definition	CONCEPTUAL	details and invariance definition filter to detect increased pulsatility as an early indicator
DO-004	deployment target + monitoring loop realizing Incorporate electrical impedance tomography sensors in definition	CONCEPTUAL	details by sensors in definition filter to detect increased pulsatility as an early indicator
DO-005	data ingestion and labeling pipeline interfaces with feature/representation layer is sufficient for graphical (time-of-flight, tolerance) electrical impedance tomography sensors	CONCEPTUAL	single energy initial (time-of-flight, tolerance) electrical impedance tomography sensors
DO-006	feature / representation layer interfaces with model + uncertainty quantification: single energy initial (time-of-flight, tolerance) electrical impedance tomography sensors	CONCEPTUAL	single energy initial (time-of-flight, tolerance) electrical impedance tomography sensors
DO-007	model + uncertainty quantification interfaces with deployment target + monitoring interface quality on (time-of-flight, tolerance) electrical impedance tomography sensors	CONCEPTUAL	single energy initial (time-of-flight, tolerance) electrical impedance tomography sensors
DO-008	Device boundaries: patient/fluid inlet and outlet termination, control/data points required by sensor, scale definition	CONCEPTUAL	control/data points required by sensor, scale definition
DO-009	Flow/force/signal path geometry realizing the mechanism to incorporate electrical impedance tomography sensors or vessels filter to detect increased pulsatility	UNKNOWN	impedance tomography sensors or vessels filter to detect increased pulsatility
DO-010	Operating range for training data volume and provenance unknown which the mechanism holds its state	UNKNOWN	is not held; computed bounds for training data volume and provenance; verification of range
DO-011	Operating range for label quality rate within which the mechanism holds its state	UNKNOWN	is not held; computed bounds for label quality rate; verification of range
DO-012	Operating range for target operating point (precision/recall) unknown which the mechanism holds its state	UNKNOWN	is not held; computed bounds for target operating point (precision/recall); verification of range
DO-013	Operating range for calibration error budget within which the mechanism holds its state	UNKNOWN	is not held; computed bounds for calibration error budget; verification of range
DO-014	Operating range for inference latency ceiling within which the mechanism holds its state	UNKNOWN	is not held; computed bounds for inference latency ceiling; verification of range
DO-015	Operating range for distribution-shift monitoring thresholds unknown which the mechanism holds its state	UNKNOWN	is not held; computed bounds for distribution-shift monitoring thresholds; verification of range
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism's operating point	PROPOSED	sensing quality, electrical impedance tomography sensors or vessels required
DO-017	Information pipeline: ingestion -> feature/representation -> model -> UQ -> decision	PROPOSED	stage input/output on loop, latency budget, missing data, threshold specification
DO-018	Fixture enabling: held-out + temporal validation splits -> unknown concept named, build specification	UNKNOWN	discrepancy, unknown selection; acceptance instrumentation
DO-019	Fixture enabling: calibration and decision-threshold analysis -> unknown concept named, build specification	UNKNOWN	discrepancy, unknown selection; acceptance instrumentation
DO-020	Fixture enabling: failure-mode stress tests (distribution shift) -> unknown concept named, build specification	UNKNOWN	discrepancy, unknown selection; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: held-out + temporal split evaluation	UNKNOWN	Unknown; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
training data volume and distribution	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
label quality rate	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
target operating point (precision)	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
calibration error budget	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
inference latency ceiling	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
distribution-shift monitoring thresholds	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
obstruction of flow	Fails by obstruction of flow: the device fails to flow at 50% of design flow (if required, designed to operate at 100% of design flow)	Design for 100% of design flow	Test at 100% of design flow	UNKNOWN (no sourced candidate) value and verification method
distribution shift degradation	Fails by distribution shift degradation: the device fails to operate at 50% of design flow (if required, designed to operate at 100% of design flow)	Design for 100% of design flow	Test at 100% of design flow	UNKNOWN (no sourced candidate) value and verification method
label noise propagating	Fails by label noise propagating: the device fails to operate at 50% of design flow (if required, designed to operate at 100% of design flow)	Design for 100% of design flow	Test at 100% of design flow	UNKNOWN (no sourced candidate) value and verification method
overfitting to source	Fails by overfitting to source: the device fails to operate at 50% of design flow (if required, designed to operate at 100% of design flow)	Design for 100% of design flow	Test at 100% of design flow	UNKNOWN (no sourced candidate) value and verification method
calibration drift invalidation	Fails by calibration drift invalidation: the device fails to operate at 50% of design flow (if required, designed to operate at 100% of design flow)	Design for 100% of design flow	Test at 100% of design flow	UNKNOWN (no sourced candidate) value and verification method
feedback-loop data corruption	Fails by feedback-loop data corruption: the device fails to operate at 50% of design flow (if required, designed to operate at 100% of design flow)	Design for 100% of design flow	Test at 100% of design flow	UNKNOWN (no sourced candidate) value and verification method

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
obstruction of flow	Fails by obstruction of flow: the device	design that not	NOT ESTABLISHED (given the high operational condition failure mode (Article X))
distribution shift degrading accuracy	Fails by distribution shift degrading accuracy	proper deployment in the target environment	On-site validation with test representative failure rate (Article X)
label noise propagating to false predictions	Fails by label noise propagating to biased predictions	systematic analysis of the data source	Use of the model-based approach to map the data (Article X)
overfitting to source-site artifacts	Fails by overfitting to source-site artifacts	the site data at acquisition	Validation with representative data (Article X)
calibration drift invalidating decisions	Fails by local threshold drift invalidating decisions	thresholds to be re-calibrated	Monitoring of the test to detect changes in the data (Article X)
feedback-loop data contamination	Fails by feedback-loop data contamination	external validation on a separate data set	Use of the model-based approach to map the data (Article X)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Compare filter obstruction rates with and without filter	Compare filter obstruction rates with and without filter	Compare filter obstruction rates with and without filter	Compare filter obstruction rates with and without filter
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria
VF-004	failure-mode stress tests (distribution shift)	failure-mode stress tests (distribution shift)	failure-mode stress tests (distribution shift)	failure-mode stress tests (distribution shift)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
NOT_APPLICABLE — data only invention: no physical materials exist to select; hardware dependencies are recorded under design	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
NOT	ESTABLISHED (no domain data established)	NOT	ESTABLISHED				

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- MLOps CI for data and model: domain registry manufacturing route - candidate only; no process qualification exists
- versioned artifact registry with rollback: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter training data volume and provenance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter label quality rate [y_i -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter target operating point (precision/recall) [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter calibration error budget [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter inference latency ceiling [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter distribution-shift monitoring thresholds [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (0.5 GPa) test	physical verification (0.5 GPa) test	ENGINEERING_PROPOSED (no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline)	first physical article	1000 hours (1000 hours)
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED (constraint compliance)	constraint-compliance envelope article	1000 hours (1000 hours)
WP-03	article for VF-001	per method	Compare filter obstruction	NOT ESTABLISHED (VF-001 verification)	article for VF-001	1000 hours (1000 hours)
WP-04	article for VF-002	per method	run killer experiment: held-out + temporal variation	NOT ESTABLISHED (VF-002 verification)	article for VF-002	1000 hours (1000 hours)
WP-05	article for VF-003	per method	held-out + temporal variation	NOT ESTABLISHED (VF-003 verification)	article for VF-003	1000 hours (1000 hours)
WP-06	article for VF-004	per method	failure-mode stress test	NOT ESTABLISHED (VF-004 verification)	article for VF-004	1000 hours (1000 hours)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.